

Remark: The utility of continuous functions is the ability to pass the limit through the function.

$$\lim_{x_n \rightarrow x} f(x_n) = f\left(\lim_{x_n \rightarrow x} x_n\right) = f(x)$$

An aside on limits. Given functions $f, g \in C$ and $c \in \mathbb{R}$, we have the following rules as $x_n \rightarrow x$:

- $\lim_{x_n \rightarrow x} (f(x_n) + g(x_n)) = \lim_{x_n \rightarrow x} f(x_n) + \lim_{x_n \rightarrow x} g(x_n)$
- $\lim_{x_n \rightarrow x} (f(x_n) \cdot g(x_n)) = \left(\lim_{x_n \rightarrow x} f(x_n)\right) \cdot \left(\lim_{x_n \rightarrow x} g(x_n)\right)$
- $\lim_{x_n \rightarrow x} (c f(x_n)) = c \cdot \lim_{x_n \rightarrow x} f(x_n)$
- If $g(x_n) \in \text{dom}(f)$, then $\lim_{x_n \rightarrow x} f(g(x_n)) = f\left(\lim_{x_n \rightarrow x} g(x_n)\right)$
- $\lim_{x_n \rightarrow x} \frac{f(x_n)}{g(x_n)}$ is defined when $\lim_{x_n \rightarrow x} g(x_n) \neq 0$

Notice: The case $\lim_{x_n \rightarrow x} g(x_n) = 0$ is indeterminate and must be handled separately.

Common strategies include:

1. Algebraic simplification
2. Squeeze Theorem
3. Asymptotic analysis
4. L'Hôpital's Rule

Theorem (Cauchy Mean Value Theorem). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Assume further that $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists $c \in (a, b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Proof. Define the auxiliary function

$$h(x) = f(x)(g(b) - g(a)) - g(x)(f(b) - f(a)).$$

Note that h is continuous on $[a, b]$ and differentiable on (a, b) since f, g are. Also observe:

$$h(a) = f(a)(g(b) - g(a)) - g(a)(f(b) - f(a)),$$

$$h(b) = f(b)(g(b) - g(a)) - g(b)(f(b) - f(a)).$$

Expanding both and simplifying shows $h(a) = h(b)$.

By Rolle's Theorem, there exists $c \in (a, b)$ such that $h'(c) = 0$. Now compute

$$h'(x) = f'(x)(g(b) - g(a)) - g'(x)(f(b) - f(a)).$$

Thus

$$0 = h'(c) = f'(c)(g(b) - g(a)) - g'(c)(f(b) - f(a)).$$

Rearranging yields

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

□

Theorem (L'Hôpital's Rule, 0/0 case). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) and continuous on $[a, b]$. Suppose

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0,$$

and that $g'(x) \neq 0$ for all $x \in (a, b)$. If the limit

$$\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

exists (finite or infinite), then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

Proof. Fix $x \in (a, b)$. Apply the Cauchy Mean Value Theorem to f and g on the interval $[a, x]$. There exists $c \in (a, x)$ such that

$$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Since $\lim_{t \rightarrow a^+} f(t) = \lim_{t \rightarrow a^+} g(t) = 0$, we have $f(a) = g(a) = 0$. Thus

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}, \quad c \in (a, x).$$

Now take the limit as $x \rightarrow a^+$. As $x \rightarrow a^+$, we also have $c \rightarrow a^+$, so

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \lim_{c \rightarrow a^+} \frac{f'(c)}{g'(c)} = L.$$

This proves the rule. □

Theorem. If f is differentiable at $a \in \mathbb{R}$, then f is continuous at a .

Proof. We must show

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Consider the difference:

$$\lim_{x \rightarrow a} (f(x) - f(a)).$$

Write this as

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot (x - a).$$

As $x \rightarrow a$, both numerator and denominator of the fraction go to 0, i.e.

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

is in a $\frac{0}{0}$ indeterminate form.

By definition of differentiability,

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a).$$

But we can also view this as a case for **L'Hôpital's Rule**, applied to

$$F(x) = f(x) - f(a), \quad G(x) = x - a.$$

Both $F(a) = 0$ and $G(a) = 0$, so by L'Hôpital:

$$\lim_{x \rightarrow a} \frac{F(x)}{G(x)} = \lim_{x \rightarrow a} \frac{F'(x)}{G'(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{1} = f'(a).$$

Therefore

$$\lim_{x \rightarrow a} (f(x) - f(a)) = \left(\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right) \cdot \lim_{x \rightarrow a} (x - a).$$

The first limit is $f'(a)$ (finite by assumption). The second limit is 0. Hence the product is $f'(a) \cdot 0 = 0$.

Thus

$$\lim_{x \rightarrow a} f(x) = f(a),$$

which proves continuity. □

Definition (Uniform Continuity). A function $f : A \rightarrow \mathbb{R}$ (with $A \subseteq \mathbb{R}$) is called **uniformly continuous** on A if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for all $x, y \in A$,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Remark. - The key difference from pointwise continuity is that in uniform continuity the δ depends only on ε (and the function), *not* on the particular point $x \in A$. - In other words, a single δ works uniformly across the whole domain.

Example. The function $f(x) = x^2$ is not uniformly continuous on the real numbers as the function tends to ∞ . However, on a bounded interval it is.

Theorem. Any continuous function on a closed bounded interval is uniformly continuous.

Definition (Partition of an interval). Let $[a, b] \subset \mathbb{R}$ be a closed interval. A **partition** P of $[a, b]$ is a finite ordered set of points

$$P = \{x_0, x_1, \dots, x_n\}, \quad a = x_0 < x_1 < \dots < x_n = b.$$

Subintervals. Each consecutive pair (x_{i-1}, x_i) is called a **subinterval**, denoted

$$[x_{i-1}, x_i], \quad i = 1, \dots, n.$$

Mesh (Norm of the Partition). The **mesh** (or norm) of the partition P is

$$\|P\| = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

Tags. A **tagged partition** is a partition $P = \{x_0, \dots, x_n\}$ together with a choice of sample point

$$t_i \in [x_{i-1}, x_i] \quad \text{for each } i = 1, \dots, n.$$

The points t_i are called **tags**.

Definition (Riemann sum). Given a tagged partition $\{([x_{i-1}, x_i], t_i)\}$ of $[a, b]$, the **Riemann sum** of $f : [a, b] \rightarrow \mathbb{R}$ is

$$S(f, P) = \sum_{i=1}^n f(t_i) (x_i - x_{i-1}).$$

Definition (Riemann integral). We say f is Riemann integrable on $[a, b]$ if there exists a real number I such that for every $\varepsilon > 0$, there exists $\delta > 0$ with the property: for any tagged partition P with $\|P\| < \delta$,

$$|S(f, P) - I| < \varepsilon.$$

In this case we write

$$I = \int_a^b f(x) dx.$$

Equally spaced partition. For $n \in \mathbb{N}$, define the partition

$$x_i = a + i\Delta x, \quad \Delta x = \frac{b-a}{n}, \quad i = 0, 1, \dots, n.$$

Thus all subintervals have equal length Δx and the mesh is $\|P\| = \Delta x$.

Choose tags as the left endpoints $t_i = x_{i-1}$. The Riemann sum becomes

$$S_n = \sum_{i=1}^n f(x_{i-1}) \Delta x.$$

Convergence to the Riemann integral. If f is continuous on $[a, b]$, then f is uniformly continuous (by the Heine–Cantor theorem). Thus for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.

Now for the equally spaced partition with $\Delta x < \delta$, we know $|x - y| < \delta$ within each subinterval. This ensures that the variation of f on each subinterval is less than ε . **Hence** the difference between any tagged Riemann sum (e.g. left endpoints, right endpoints, or midpoints) and the integral is bounded by a term that goes to 0 as $n \rightarrow \infty$.

Therefore,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_{i-1}) \Delta x = \int_a^b f(x) dx.$$

Motivation: What is the *computer representation of a number*?

Number systems

Defn: A **base** is the number used as the reference frame for constructing a number system.

e.g. Base-10: 0, 1, 2, 3, ..., 9

$$\text{ex: } 5239 = 5 \times 10^3 + 2 \times 10^2 + 3 \times 10^1 + 9 \times 10^0$$

e.g. Base-2 (**binary**): 0, 1

ex: $(1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

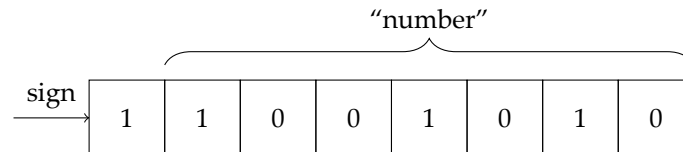
Each binary digit is referred to as a **bit**.

Binary representation of integers

Suppose we have 8 bits to represent an integer $z \in \mathbb{Z}$.

sign bit = 0 $\rightarrow$ positive integer

sign bit = 1 $\rightarrow$ negative integer



Ex: $(11001010)_2 = (-1) \times (1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0) = -(74)_{10}$

Notice: the 8 bit representation has a range of $(-127, 127)$.

Prove by induction: the n bit representation has a maximum positive value of $2^{n-1} - 1$.

Binary representation of floating point numbers

Motivation: What if we want to include decimal representations?

e.g. $(62.54)_{10} = 6 \times 10^1 + 2 \times 10^0 + 5 \times 10^{-1} + 4 \times 10^{-2}$

e.g. $(11.01)_2 = 1 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} = 3.25$

Canonical Form

Notice that the decimal can be placed anywhere $(62.54)_{10} = 0.6254 \times 10^2 = 624500 \times 10^{-4}$

So we introduce a **canonical form**: 6.254×10^{-1} – one non-zero entry left of the decimal. (i.e. **scientific notation**)

General Form: In general, any $x \in \mathbb{R}$ can be written as

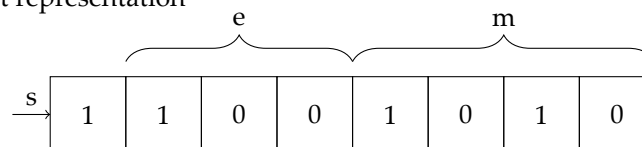
$$x = -1^s \cdot m \cdot b^e$$

where

- s : sign
- m : mantissa
- b : base
- e : exponent

Using the canonical form we take m to be normalized as $1.f$ (e.g. $*. * \dots \times 2^e$ where $*$ is a non-zero value)

Using 8-bit floating point representation



Ex: $(11001010)_{8\text{-bit float}} = (-)(1.1010 \times 2^4) = (-)(1 \times 2^4 + 1 \times 2^3 + 1 \times 2^1) = -(26)_{10}$

Floating point errors

Motivation: Machines represent $x \in \mathbb{R}$ within a certain level of precision using floating point numbers.

$$\text{True Value} = \text{Approximate Value} + \text{Error}$$

Defn: Machine precision, $\varepsilon_{\text{mach}}$ is the smallest $\varepsilon \in \mathbb{R}$ with $\varepsilon > 0$ such that $\text{fl}(1 + \varepsilon) > \text{fl}(1)$ where fl denotes the floating point representation of the number.

Floating point operations (flops)

Stability

Let $\mathbf{e}^0$ denote the initial error and $\mathbf{e}^n$ the error after n steps. A scheme is *stable* if there exists a constant C independent of n and of the input data such that

$$\|\mathbf{e}^n\| \leq C \|\mathbf{e}^0\| \quad \text{for all } n,$$

where h is the step size (or, more generally, a discretization parameter). This is *uniform boundedness* of the error amplification.

Remark: More generally, many error analyses yield

$$\|\mathbf{e}^n\| \leq C(n) \|\mathbf{e}^0\|,$$

for some function $C(n)$. Typical regimes:

- $C(n) \equiv C$ (uniform bound): errors remain uniformly controlled (unconditional stability on the given horizon).
- $C(n) = C^n$ with $C < 1$: geometric decay; with $C = 1$: bounded; with $C > 1$: geometric growth.

Remark (loss of uniformity). If the bound depends on discretization parameters (e.g., $C = C(h)$) or grows with n unless a step-size condition holds, then stability is *conditional*: the method is stable only when those conditions (CFL-type, spectral, etc.) are enforced. If no such bound exists, the method is unstable.

Linear iteration example. Consider the linear update

$$\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n, \quad \tilde{\mathbf{x}}_0 = \mathbf{x}_0 + \delta,$$

so the error satisfies $\mathbf{e}^n = \mathbf{A}^n \delta$ and

$$\|\mathbf{e}^n\| \leq \|\mathbf{A}^n\| \|\delta\|.$$

From submultiplicativity, $\|\mathbf{A}^n\| \leq \|\mathbf{A}\|^n$, hence

$$\|\mathbf{e}^n\| \leq \|\mathbf{A}\|^n \|\delta\|.$$

Consequences (in any induced operator norm):

- If $\|\mathbf{A}\| < 1$, then $\|\mathbf{e}^n\| \leq \|\mathbf{A}\|^n \|\delta\| \rightarrow 0$ (geometric decay).
- If $\|\mathbf{A}\| = 1$, then $\|\mathbf{e}^n\| \leq \|\delta\|$ (uniformly bounded).
- If $\|\mathbf{A}\| > 1$, the bound grows like $\|\mathbf{A}\|^n$ (at best), so errors can amplify.

A sharper spectral statement: $\mathbf{A}^n \rightarrow \mathbf{0}$ iff the spectral radius $\rho(\mathbf{A}) < 1$. For *normal* $\mathbf{A}$ (e.g., symmetric), $\|\mathbf{A}\|_2 = \rho(\mathbf{A})$, and the norm and spectral criteria coincide.

Takeaway. The goal of this class will be to analyze: (i) the error $\mathbf{e}^n$ and its amplification, and (ii) the bounds C (or $C(n)$). Uniform bounds yield unconditional stability; bounds that require parameter relations yield conditional stability.

2.1 Root Finding

How to guarantee a solution exists? i.e. how to know $\exists x \in \text{dom}(f)$ such that $f(x) = 0$.

Due to the intermediate value theorem.

Note: IVT is proved by taking $s = \{x | f(x) \leq 0\}$ then for $f \in C[a, b]$, at least $a \in s$. But because f is continuous then for some $\epsilon > 0$ then $f(a + \epsilon) \in s$ as long as $a \neq \sup(s)$. Thus $c = \sup(s)$.

How to guarantee a unique solution exists? $\exists! x \in \text{dom}(f)$ such that $f(x) = 0$? – use the derivative.

Not universal but one common property is monotonicity $f \in C^1[a, b]$ and $f'(x) > 0$ for all $x \in [a, b]$.

Bisection Algorithm

Setup. Let f be continuous on $[a, b]$ with $f(a)f(b) < 0$. (Assume a unique root $p \in (a, b)$.)

Iteration. Initialize $a_1 = a$, $b_1 = b$. For $n \geq 1$:

$$p_n = a_n + \frac{b_n - a_n}{2} = \frac{a_n + b_n}{2}.$$

(note the first expression more numerically stable).

$$\begin{cases} \text{if } f(p_n) = 0 \text{ then stop (root found);} \\ \text{if } f(a_n)f(p_n) < 0 \Rightarrow a_{n+1} = a_n, b_{n+1} = p_n; \\ \text{else } a_{n+1} = p_n, b_{n+1} = b_n. \end{cases}$$

Stopping criteria. For a tolerance $\epsilon > 0$, terminate at the smallest N such that

$$\begin{aligned} |p_N - p_{N-1}| &< \epsilon, \\ \frac{|p_N - p_{N-1}|}{|p_N|} &< \epsilon \quad (p_N \neq 0), \\ |f(p_N)| &< \epsilon. \end{aligned}$$

Analyzing the Bisection Algorithm

Theorem 1. Suppose $f \in C[a, b]$ and $f(a)f(b) < 0$. The Bisection method generates a sequence $\{p_n\}_{n=1}^{\infty}$ approximating a zero p of f with

$$|p_n - p| \leq \frac{b - a}{2^n}, \quad n \geq 1.$$

Proof. For each $n \geq 1$, we have

$$b_n - a_n = \frac{1}{2^{n-1}} (b - a)$$

and

$$p \in (a_n, b_n).$$

Since $p_n = \frac{1}{2}(a_n + b_n)$ for all $n \geq 1$, it follows that

$$|p_n - p| \leq \frac{1}{2} (b_n - a_n) = \frac{b - a}{2^n}.$$

□

Note: this is only a bound for the error and may be quite a conservative bound.

Fixed Points

Definition 1 (Fixed point). A number p is a fixed point of a function g if $g(p) = p$.

Equivalence with root-finding. Root-finding $f(p) = 0$ and fixed-point problems $g(p) = p$ are interchangeable:

- Given $f(p) = 0$, define g so that p is a fixed point, e.g.

$$g(x) = x - f(x) \quad \text{or} \quad g(x) = x + 3f(x).$$

Then $g(p) = p$ whenever $f(p) = 0$.

- Conversely, if g has a fixed point p ($g(p) = p$), then

$$f(x) = x - g(x)$$

satisfies $f(p) = 0$.

Remarks. Although our goal is often to solve $f(x) = 0$, the fixed-point form can be easier to analyze, and suitable choices of g lead to powerful root-finding schemes.

Example. Determine the fixed points of $g(x) = x^2 - x$.

Solution. A fixed point p satisfies

$$p = g(p) = p^2 - p \implies 0 = p^2 - 2p = p(p - 2).$$

Hence the fixed points are

$$p = 0 \quad \text{and} \quad p = 2,$$

i.e., the intersections of $y = g(x)$ with $y = x$.

Theorem 2 (Fixed Point Existence and Uniqueness). (i) If $g \in C[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$, then g has at least one fixed point in $[a, b]$.

(ii) If, in addition, $g'(x)$ exists on (a, b) and there is a constant $k < 1$ such that

$$|g'(x)| \leq k \quad \text{for all } x \in (a, b),$$

then there is exactly one fixed point in $[a, b]$.

Proof. (i) If $g(a) = a$ or $g(b) = b$, then g has a fixed point at an endpoint. Otherwise $g(a) > a$ and $g(b) < b$. Define $h(x) = g(x) - x$. Then h is continuous on $[a, b]$, with

$$h(a) = g(a) - a > 0, \quad h(b) = g(b) - b < 0.$$

By the Intermediate Value Theorem there exists $p \in (a, b)$ such that $h(p) = 0$. Hence $0 = h(p) = g(p) - p$, i.e., $g(p) = p$.

(ii) Assume in addition that $|g'(x)| \leq k < 1$ on (a, b) and suppose p and q are both fixed points in $[a, b]$. If $p \neq q$, then by the Mean Value Theorem there exists ξ between p and q (hence $\xi \in (a, b)$) such that

$$\frac{g(p) - g(q)}{p - q} = g'(\xi).$$

Therefore

$$|p - q| = |g(p) - g(q)| = |g'(\xi)| |p - q| \leq k |p - q| < |p - q|,$$

a contradiction. Thus $p = q$, and the fixed point in $[a, b]$ is unique. $\square$

Highlight: IVT and MVT (as they appear many times). Highlight how derivative conditions are very useful for ensuring uniqueness as we've also seen when discussing root finding.

Example 2 Show that $g(x) = \frac{x^2 - 1}{3}$ has a unique fixed point on the interval $[-1, 1]$.

Solution. The extreme values of g on $[-1, 1]$ occur at endpoints or where $g'(x) = 0$. Since $g'(x) = \frac{2x}{3}$, g is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$, with critical point $x = 0$ and endpoints $x = \pm 1$. Compute

$$g(-1) = 0, \quad g(1) = 0, \quad g(0) = -\frac{1}{3}.$$

Hence $g([-1, 1]) \subset [-\frac{1}{3}, 0] \subset [-1, 1]$.

Moreover,

$$|g'(x)| = \left| \frac{2x}{3} \right| \leq \frac{2}{3} < 1 \quad \text{for all } x \in (-1, 1).$$

Thus g satisfies all hypotheses of Theorem 2: it maps $[-1, 1]$ into itself and $|g'| \leq k < 1$ on $(-1, 1)$. Therefore g has *exactly one* fixed point in $[-1, 1]$.

Fixed Point Approximation

To approximate a fixed point of a function g , choose an initial guess p_0 and generate the sequence $\{p_n\}_{n=0}^{\infty}$ by

$$p_n = g(p_{n-1}), \quad n \geq 1.$$

If the sequence converges to p and g is continuous, then

$$p = \lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} g(p_{n-1}) = g\left(\lim_{n \rightarrow \infty} p_{n-1}\right) = g(p),$$

so p solves $x = g(x)$. This procedure is called *fixed-point* (or *functional*) iteration.

Algorithm 1 Fixed-Point (Functional) Iteration for $p = g(p)$ **Require:** function g , initial guess p_0 , tolerance TOL, max iterations $N_{\max}$ **Ensure:** approximate fixed point p or failure

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1:  $i \leftarrow 1$ 
2: while  $i \leq N_{\max}$  do
3:    $p \leftarrow g(p_0)$  ▷ compute  $p_i$ 
4:   if  $|p - p_0| < \text{TOL}$  then
5:     return  $p$  ▷ success
6:   end if
7:    $p_0 \leftarrow p$  ▷ update iterate
8:    $i \leftarrow i + 1$ 
9: end while
10: return failure: no convergence within  $N_{\max}$  iterations

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Theorem 3 (Convergence of Fixed-Point Iteration). Let $g \in C[a, b]$ and suppose $g(x) \in [a, b]$ for all $x \in [a, b]$. Assume, in addition, that g' exists on (a, b) and there is a constant $0 < k < 1$ with

$$|g'(x)| \leq k, \quad \text{for all } x \in (a, b).$$

Then, for any $p_0 \in [a, b]$, the sequence defined by

$$p_n = g(p_{n-1}), \quad n \geq 1,$$

converges to the unique fixed point $p \in [a, b]$.

Proof. By Theorem 2, there exists a unique $p \in [a, b]$ with $g(p) = p$. Since g maps $[a, b]$ into itself, the sequence $\{p_n\}_{n=0}^{\infty}$ is well-defined and $p_n \in [a, b]$ for all n . Using $|g'(x)| \leq k$ and the Mean Value Theorem, for each n there exists $\xi_n \in (a, b)$ such that

$$|p_n - p| = |g(p_{n-1}) - g(p)| = |g'(\xi_n)| |p_{n-1} - p| \leq k |p_{n-1} - p|.$$

Applying this inductively yields

$$|p_n - p| \leq k |p_{n-1} - p| \leq k^2 |p_{n-2} - p| \leq \cdots \leq k^n |p_0 - p|. \quad (2.4)$$

Since $0 < k < 1$, we have $\lim_{n \rightarrow \infty} k^n = 0$, hence

$$\lim_{n \rightarrow \infty} |p_n - p| \leq \lim_{n \rightarrow \infty} k^n |p_0 - p| = 0.$$

Therefore $p_n \rightarrow p$, as claimed. □

If g satisfies the hypotheses of Theorem 2 and the convergence theorem above (i.e., $g : [a, b] \rightarrow [a, b]$, $g \in C[a, b]$, and $|g'| \leq k < 1$ on (a, b)), then the errors of fixed-point iteration $p_n = g(p_{n-1})$ satisfy

$$|p_n - p| \leq k^n \max\{p_0 - a, b - p_0\}, \quad (2.5)$$

and, for all $n \geq 1$,

$$|p_n - p| \leq \frac{k^n}{1 - k} |p_1 - p_0|. \quad (2.6)$$

Take-away. How can we choose a fixed-point formulation that reliably and rapidly converges to the solution of a given root-finding problem?

Answer. Rewrite the equation $f(x) = 0$ as $x = g(x)$ so that g maps $[a, b]$ into itself and satisfies the hypotheses of the fixed-point theorem (continuous, $|g'(x)| \leq k < 1$ near the solution). Aim to make $|g'(p)|$ as small as possible to accelerate convergence.

In the next sections we examine constructions of such g in detail.

Newton's Method and Its Extensions

Assume $f \in C^2[a, b]$. Let $p_0 \in [a, b]$ be an approximation to a root p with $f'(p_0) \neq 0$ and $|p - p_0|$ small. The first-order Taylor expansion of f about p_0 , evaluated at $x = p$, gives

$$f(p) = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p)),$$

where $\xi(p)$ lies between p and p_0 . Since $f(p) = 0$,

$$0 = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p)).$$

If $|p - p_0|$ is small, the quadratic term is much smaller, so

$$0 \approx f(p_0) + (p - p_0)f'(p_0).$$

Solving for p yields the Newton update

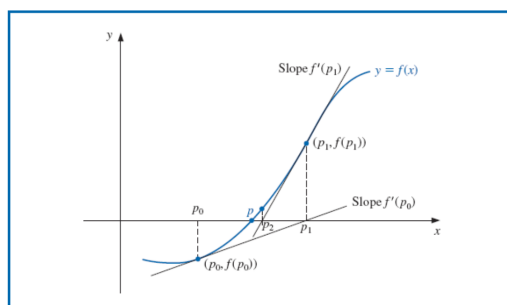
$$p \approx p_0 - \frac{f(p_0)}{f'(p_0)} \equiv p_1.$$

Iteration. Starting from p_0 , define $\{p_n\}_{n=0}^\infty$ by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n \geq 1. \quad (2.7)$$

Figure 2.7 illustrates how the approximations are obtained using successive tangents. (Also see Exercise 31.) Starting with the initial approximation p_0 , the approximation p_1 is the x -intercept of the tangent line to the graph of f at $(p_0, f(p_0))$. The approximation p_2 is the x -intercept of the tangent line to the graph of f at $(p_1, f(p_1))$ and so on. Algorithm 2.3 implements this procedure.

Figure 2.7



► Details

Algorithm 2 Newton's Method**Require:** function f , derivative f' , initial guess p_0 , tolerance TOL, max iterations $N_{\max}$ **Ensure:** approximate root p or failure

```

1:  $i \leftarrow 1$ 
2: while  $i \leq N_{\max}$  do
3:    $p \leftarrow p_0 - \frac{f(p_0)}{f'(p_0)}$  ▷ Newton update
4:   if  $|p - p_0| < \text{TOL}$  or  $|f(p)| < \text{TOL}$  then
5:     return  $p$  ▷ success
6:   end if
7:    $p_0 \leftarrow p$ 
8:    $i \leftarrow i + 1$ 
9: end while
10: return failure: no convergence within  $N_{\max}$  iterations

```

Theorem 4 (Local convergence of Newton's method). Let $f \in C^2[a, b]$. If $p \in (a, b)$ satisfies $f(p) = 0$ and $f'(p) \neq 0$, then there exists $\delta > 0$ such that Newton's method generates a sequence $\{p_n\}_{n=1}^{\infty}$ converging to p for any initial approximation $p_0 \in [p - \delta, p + \delta]$.

Proof. View Newton's method as the functional iteration $p_n = g(p_{n-1})$ with

$$g(x) = x - \frac{f(x)}{f'(x)}, \quad n \geq 1.$$

Choose $k \in (0, 1)$. Since f' is continuous and $f'(p) \neq 0$, there exists $\delta_1 > 0$ such that $f'(x) \neq 0$ for $x \in [p - \delta_1, p + \delta_1] \subset (a, b)$. Hence g is defined and continuous on $[p - \delta_1, p + \delta_1]$. Moreover,

$$g'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2},$$

so, using $f(p) = 0$,

$$g'(p) = \frac{f(p)f''(p)}{[f'(p)]^2} = 0.$$

Because g' is continuous and $0 < k < 1$, there exists δ with $0 < \delta < \delta_1$ such that

$$|g'(x)| \leq k, \quad \text{for all } x \in [p - \delta, p + \delta].$$

It remains to show that g maps $[p - \delta, p + \delta]$ into itself. For $x \in [p - \delta, p + \delta]$, the Mean Value Theorem gives a ξ between x and p with

$$|g(x) - p| = |g(x) - g(p)| = |g'(\xi)| |x - p| \leq k |x - p| < |x - p|.$$

Since $x \in [p - \delta, p + \delta]$, we have $|x - p| < \delta$, hence $|g(x) - p| < \delta$, so $g([p - \delta, p + \delta]) \subset [p - \delta, p + \delta]$.

All hypotheses of the fixed-point contraction theorem now hold; therefore, the iteration

$$p_n = g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n \geq 1,$$

converges to p for any $p_0 \in [p - \delta, p + \delta]$. □

Newton's Method Assume $f \in C^2[a, b]$. Let $p_0 \in [a, b]$ be an approximation to a root p with $f'(p_0) \neq 0$ and $|p - p_0|$ small. The first-order Taylor expansion of f about p_0 and solving for p , gives the Newton update.

Last time we saw that if the initial guess p_0 is sufficiently close to p then the method will converge. Today we will answer – how close does it need to be.

The takeaway will be that similar to how first derivatives controlled behavior of bisection methods, now for the second-order Newton's method the first and second-order derivatives will be used to determine "how close".

Theorem 5 (Newton step and local error). Assume $f \in C^2[a, b]$ and $p \in [a, b]$ is a simple root: $f(p) = 0$, $f'(p) \neq 0$. For p_0 sufficiently close to p , define the Newton iterates

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}.$$

Then for each n there exists ξ_n between p_n and p such that

$$p - p_{n+1} = -\frac{f''(\xi_n)}{2f'(p_n)}(p - p_n)^2.$$

This follows directly from the Taylor series expansion.

Theorem 6 (Quadratic convergence of Newton's method). Assume $f \in C^2[a, b]$ and let $p \in (a, b)$ satisfy $f(p) = 0$ and $f'(p) \neq 0$. If p_0 is sufficiently close to p , the Newton iterates

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}, \quad n \geq 0,$$

are well defined, satisfy $p_n \rightarrow p$, and

$$\lim_{n \rightarrow \infty} \frac{p - p_{n+1}}{(p - p_n)^2} = -\frac{f''(p)}{2f'(p)}.$$

In particular, the order of convergence is $p = 2$.

Proof. Since $f'(p) \neq 0$ and f' is continuous, choose $\varepsilon > 0$ with $I = [p - \varepsilon, p + \varepsilon] \subset (a, b)$ and $\min_{x \in I} |f'(x)| > 0$. Define

$$M := \frac{\max_{x \in I} |f''(x)|}{2 \min_{x \in I} |f'(x)|}.$$

By Taylor's theorem with Lagrange remainder, for $p_n \in I$ there exists ξ_n between p_n and p such that

$$p - p_{n+1} = -\frac{f''(\xi_n)}{2f'(p_n)}(p - p_n)^2.$$

Taking absolute values and using the definition of M yields the key estimate

$$|p - p_{n+1}| \leq M |p - p_n|^2. \quad (1)$$

Staying in I and convergence. Pick $p_0 \in I$ so close to p that $|p - p_0| \leq \varepsilon$ and $M|p - p_0| < 1$. Then (1) gives $M|p - p_1| \leq (M|p - p_0|)^2 < M|p - p_0| \leq 1$, hence $|p - p_1| \leq \varepsilon$; inductively, $p_n \in I$ for all n , and

$$M|p - p_n| \leq (M|p - p_0|)^{2^n}.$$

Thus $|p - p_n| \rightarrow 0$, i.e. $p_n \rightarrow p$.

Asymptotic constant. Because $p_n \rightarrow p$ and ξ_n lies between p_n and p , we have $\xi_n \rightarrow p$. Continuity of f' and f'' then gives $f'(p_n) \rightarrow f'(p)$ and $f''(\xi_n) \rightarrow f''(p)$. Divide the exact error identity by $(p - p_n)^2$ to obtain

$$\frac{p - p_{n+1}}{(p - p_n)^2} = -\frac{f''(\xi_n)}{2f'(p_n)} \rightarrow -\frac{f''(p)}{2f'(p)}.$$

This proves quadratic convergence with the stated limit. $\square$

Lemma 1 (**Practical convergence test for Newton**). Let $f \in C^2[a, b]$ and $p \in (a, b)$ with $f(p) = 0$, $f'(p) \neq 0$. Choose $I = [p - \varepsilon, p + \varepsilon] \subset (a, b)$ so that

$$m := \min_{x \in I} |f'(x)| > 0, \quad M_2 := \max_{x \in I} |f''(x)| < \infty,$$

and set

$$M := \frac{M_2}{2m}.$$

If the initial guess $p_0 \in I$ satisfies

$$|p - p_0| < \frac{1}{M},$$

then the Newton iterates $p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}$ stay in I and converge to p , with

$$|p - p_{n+1}| \leq M |p - p_n|^2.$$

Interpretation: larger M (large curvature or small slope) requires a closer initial guess.

The Secant Method

Newton's method is powerful but requires evaluating f' at each step. To avoid derivative evaluations, approximate f' by a secant slope. By definition,

$$f'(p_{n-1}) = \lim_{x \rightarrow p_{n-1}} \frac{f(x) - f(p_{n-1})}{x - p_{n-1}}.$$

If p_{n-2} is close to p_{n-1} , then

$$f'(p_{n-1}) \approx \frac{f(p_{n-1}) - f(p_{n-2})}{p_{n-1} - p_{n-2}}.$$

Definition. Substituting this in Newton's update yields the *secant* iteration

$$p_n = p_{n-1} - \frac{f(p_{n-1})(p_{n-1} - p_{n-2})}{f(p_{n-1}) - f(p_{n-2})}, \quad n \geq 2. \quad (2)$$

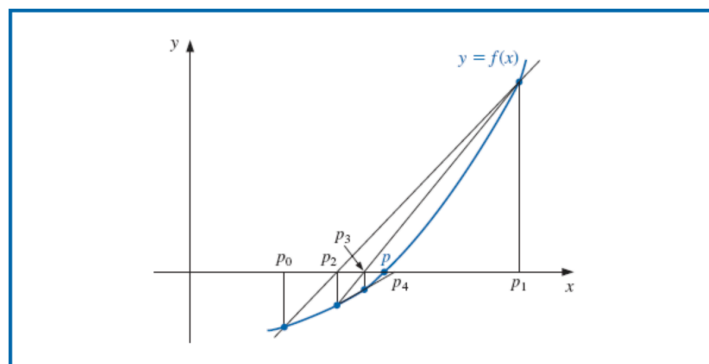
Geometric interpretation. Starting from two initial approximations p_0 and p_1 :

- p_2 is the x -intercept of the line through $(p_0, f(p_0))$ and $(p_1, f(p_1))$.
- p_3 is the x -intercept of the line through $(p_1, f(p_1))$ and $(p_2, f(p_2))$, etc.

The Method of False Position

Idea. Like the Secant method, use the secant line to estimate the root, but *always keep the root bracketed*. Start with a, b such that $f(a)f(b) < 0$. At each step form the secant through $(a, f(a))$ and $(b, f(b))$ and take

Figure 2.9



▼ Details

A curve is graphed on an x y coordinate plane. The x axis has tick marks labeled from the left to the right as follows: p_0 , p_2 , p_3 , p_4 , and p_1 . 5 points are plotted, in which a point each is plotted directly below p_0 , p_2 , and p_3 , respectively. A point p is marked on the x axis between p_3 and p_4 , close to p_4 . The fifth point is marked directly above p_1 . The curve labeled $y = f(x)$ rises as a concave up increasing curve from the fourth quadrant to the first quadrant through the 5 points. Four vertical lines are drawn from p_0 , p_2 , p_3 , and p_1 on the x axis to the curve. Two upward sloping lines are drawn from the plotted points below $x = p_0$ and p_2 in the fourth quadrant to the plotted point above $x = p_1$ in the first quadrant through p_2 and p_3 on the x axis, respectively. A small line slopes upward from the plotted point below $x = p_2$ to the point p_4 on the x axis.

its x -intercept as the next approximation. Replace the endpoint that has the *same* sign as the new value so that the sign change (and hence the bracket) is preserved.

Secant intercept. Given current bracket $[a, b]$ with $f(a)f(b) < 0$, define

$$p = b - \frac{f(b)(b-a)}{f(b)-f(a)} = a - \frac{f(a)(a-b)}{f(a)-f(b)}.$$

Bracketing update.

If $f(a)f(p) < 0$ then $b \leftarrow p$; else $a \leftarrow p$.

This guarantees $f(a)f(b) < 0$ after every iteration.

Algorithm 3 False Position

Require: function f , initial a, b with $f(a)f(b) < 0$, tolerance TOL, max its $N_{\max}$

Ensure: approximate root p (bracket maintained) or failure

```

1: for  $n = 1$  to  $N_{\max}$  do
2:    $p \leftarrow b - \frac{f(b)(b-a)}{f(b)-f(a)}$  ▷  $x$ -intercept of secant
3:   if  $|f(p)| < \text{TOL}$  or  $\max(|p-a|, |b-p|) < \text{TOL}$  then
4:     return  $p$ 
5:   end if
6:   if  $f(a)f(p) < 0$  then
7:      $b \leftarrow p$ 
8:   else
9:      $a \leftarrow p$ 
10:  end if
11: end for
12: return failure: no convergence within  $N_{\max}$  iterations

```

Remarks.

- Each step needs only one new function evaluation (after the first two).
- Bracketing is preserved, so a root remains in $[a, b]$; however, unlike bisection, the interval length need not halve each step.

Theorem 7 (Steffensen's method is quadratic (limit p)). Let $g \in C^2$ in a neighborhood of a fixed point p with $g(p) = p$ and $\lambda := g'(p)$ satisfying $|\lambda| < 1$ and $\lambda \neq 1$. Define Steffensen's iteration

$$x_{n+1} = x_n - \frac{(g(x_n) - x_n)^2}{g(g(x_n)) - 2g(x_n) + x_n}.$$

For x_0 sufficiently close to p , the iterates $x_n \rightarrow p$ and

$$x_{n+1} - p = \frac{g''(p)}{2(1-\lambda)} (x_n - p)^2 + o((x_n - p)^2),$$

i.e. the convergence is quadratic.

Sketch. Let $e_n := x_n - p$. Taylor at p :

$$g(p + e) = p + \lambda e + ce^2 + o(e^2), \quad c := \frac{1}{2}g''(p).$$

Then $g(x_n) - x_n = (\lambda - 1)e_n + ce_n^2 + o(e_n^2)$ and $g(g(x_n)) = p + \lambda^2 e_n + c(\lambda + \lambda^2)e_n^2 + o(e_n^2)$. Hence

$$D := g(g(x_n)) - 2g(x_n) + x_n = (1 - \lambda)^2 e_n + O(e_n^2), \quad N := (g(x_n) - x_n)^2 = (1 - \lambda)^2 e_n^2 + O(e_n^3).$$

So

$$\frac{N}{D} = e_n - \frac{2c}{1 - \lambda} e_n^2 + o(e_n^2),$$

and

$$e_{n+1} = e_n - \frac{N}{D} = \frac{2c}{1 - \lambda} e_n^2 + o(e_n^2) = \frac{g''(p)}{2(1 - \lambda)} e_n^2 + o(e_n^2).$$

□

Theorem 8 (Steffensen's method is quadratic). Let $g \in C^2$ in a neighborhood of a fixed point α with $g(\alpha) = \alpha$ and $\lambda := g'(\alpha)$ satisfying $|\lambda| < 1$ and $\lambda \neq 1$. Define Steffensen's iteration

$$x_{n+1} = x_n - \frac{(g(x_n) - x_n)^2}{g(g(x_n)) - 2g(x_n) + x_n}.$$

Then for x_0 sufficiently close to α , the iterates $x_n \rightarrow \alpha$ and

$$x_{n+1} - \alpha = \frac{g''(\alpha)}{2(1 - \lambda)} (x_n - \alpha)^2 + o((x_n - \alpha)^2),$$

i.e. the convergence is quadratic.

Sketch. Write $e_n := x_n - \alpha$. Taylor-expand g at α :

$$g(\alpha + e) = \alpha + \lambda e + ce^2 + o(e^2), \quad c := \frac{1}{2}g''(\alpha).$$

Hence

$$g(x_n) - x_n = (\lambda - 1)e_n + ce_n^2 + o(e_n^2).$$

Apply g again with $\delta := \lambda e_n + ce_n^2 + o(e_n^2)$:

$$g(g(x_n)) = \alpha + \lambda \delta + c\delta^2 + o(e_n^2) = \alpha + \lambda^2 e_n + c(\lambda + \lambda^2)e_n^2 + o(e_n^2).$$

Therefore the denominator

$$D := g(g(x_n)) - 2g(x_n) + x_n = (1 - \lambda)^2 e_n + O(e_n^2),$$

and the numerator

$$N := (g(x_n) - x_n)^2 = (1 - \lambda)^2 e_n^2 + O(e_n^3).$$

Thus

$$\frac{N}{D} = \frac{(1 - \lambda)^2 e_n^2 + O(e_n^3)}{(1 - \lambda)^2 e_n + O(e_n^2)} = e_n - \frac{2c}{1 - \lambda} e_n^2 + o(e_n^2).$$

Finally,

$$e_{n+1} = x_{n+1} - \alpha = e_n - \frac{N}{D} = \frac{2c}{1 - \lambda} e_n^2 + o(e_n^2) = \frac{g''(\alpha)}{2(1 - \lambda)} e_n^2 + o(e_n^2),$$

which proves quadratic convergence. □